Discovering Patterns of Touch

A Case Study for Visualization-Driven Analysis in Human-Robot Interaction

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***Abstract*— An important challenge in Human-Robot Interaction (HRI) is the analysis and interpretation of large volumes of data to inform the design process within a complex, multi-faceted research space. In this study, we explore how data visualization techniques can contribute to HRI methodology, particularly in terms of linking qualitative and quantitative analysis methods. Specifically, we present a case study demonstrating the visualization of touch data to identify potential patterns of interaction with a social robot intended for deployment in preschool classrooms.**

Keywords*— *Human-robot interaction (HRI); Data visualization; Data analysis; Parallel Sets; Patterns of touch; OPAL Robot

I. Introduction

An important issue that emerges in HRI contexts is the need to define a principled approach to guide the overarching research and development process. An important factor in this regard is the vast research space that results from numerous potential design and experimental strategies. Accordingly, exhaustive investigation is rarely feasible and creative, mixed method research approaches are often advantageous [1].

A review of mixed method research approaches in practice found some of the most significant benefits to be: mutual corroboration; illustration and enhancement of clarity, and; extended scope of inquiry [2]. Another important observation in the review was the difference between the rationale expressed for using a mixed method approach and the actual benefits that were noted. Of particular interest in this paper is how the process of interpreting results can provide unanticipated insights and cross-method corroboration. Following on, we investigate how such benefits might be intentionally sought through the utilization of data visualization techniques.

One of the most valuable aspects of visualization is the capacity to distill information, limiting the need to manually examine large quantities of raw data [3]. This characteristic is particularly relevant in the HRI domain where vast amounts of data are generated by sensors and from both automatic and manual coding of video recordings. Consequently, visualizations can provide increased accessibility for analysts, allowing them to rapidly draw inferences without the need for a well-established viewpoint or hypothesis. The ability to derive insights via qualitative exploration of data abstractions can be extremely powerful, quickly revealing high-opportunity topics for further, quantitative analysis. This relationship between analysis methods can be employed in an evolving, iterative context with each successive application of a particular technique helping to inform and refine subsequent analyses.

II. Case Study: Patterns of Touch

A. Data Collection

For this investigation, we evaluated manually coded touch data for video recordings of interactions between school children (aged one to six years old) and a social robot named Opie [4] intended for deployment in preschool classrooms. The interactions were carried out as part of an early prototyping phase [5] and took place at a public technology event, with the children playing between one and four games[1]. Three games were played exclusively on a tablet located in the robot's torso and one game was played through verbal interaction. GoPro recordings were made of all interaction sessions and subsequent manual coding produced a log of all noticeable touches of the robot made by the children. Touches were categorised by their origin (prompted and spontaneous) and general location (hands, torso, and other). For further details see [4] [6].

B. Data Visualisations

The data was then visualized using a range of methods, including temporal raster plots that show touch sequences for each child (see [6]), and a technique for exposing relationships between categorical variables known as Parallel Sets, which was originally developed as an abstraction of parallel coordinates for working with high-dimensional survey data [7]. Our visualizations were implemented in JavaScript using the D3.js framework[2, 3]. The predominant motivation for investigating touch data with Parallel Sets was to reveal potential patterns of touch for multi-participant interactions with the robot. For this investigation we examined relationships between three variables in visualizations of a total of six interaction sessions. Visualizations of two of these interaction sessions are presented in Fig. 1.

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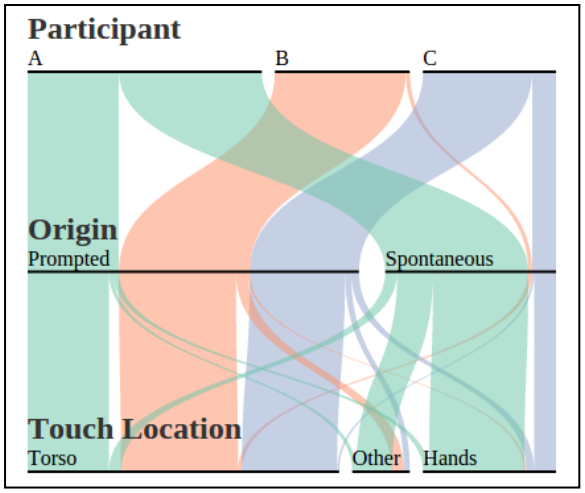
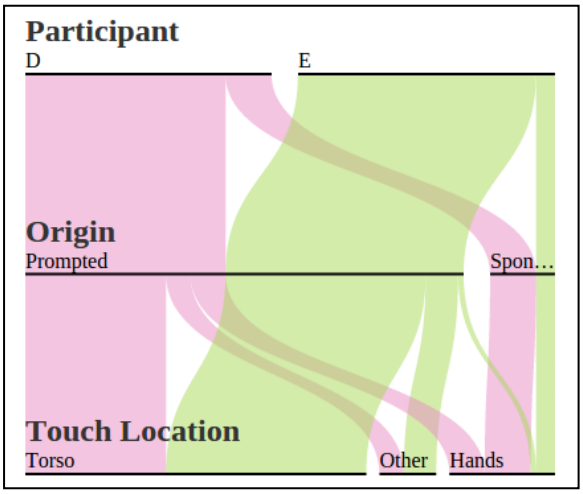
[1] Ethical clearance: Title – "RUBI-Oz Design Pilot Test" clearance number: 15-PSYCH-PHD-42-JH

[2] http://d3js.org/

[3] https://github.com/jasondavies/d3-parsets



(a) Session 1 – Three children (A, B, C)



(b) Session 2 – Two children (D, E)

Fig. 1 – Parallel Sets visualizations showing activity ratios between Participants, Origin and Touch Location in two interaction sessions.

C. *Analysis of Relationships between Categorical Variables*

Parallel Sets may be difficult to interpret on first impression but their key aspects become clear with experience. They are read as a flow from top to bottom, with points of divergence showing relative proportions for each categorical variable. More specifically, the width of each variable flow represents its relative contribution towards the total number of touches made by all participants in that interaction session. Interactive reordering of variables enables dynamic exploration of relationships. In our case, an analyst with experience in similar visualization techniques was able to draw three distinct and significant inferences without cross-referencing the raw data or video recordings.

Firstly, one of the most apparent relationships was the bias of spontaneous touching towards the hands of the robot, which is illustrated by the outflow proportions for "Spontaneous" in Fig. 1(a) and "Spon…" in Fig. 1(b). The same tendency was also observed in visualizations of other sessions not included in this report. Further investigation could test for the role of hand touching as a potential social cue in human-robot interactions.

Secondly, in a related observation, there appears to be a distinction between types of spontaneous touching behaviors. Specifically, the spontaneous touches of some children (C, D, E) are directed almost exclusively towards the hands, whereas participant A shows significant spontaneous touching of the torso and other regions. Future testing could evaluate the presence of different touch tendencies between children.

Finally, and illustrated by the contrast between Fig. 1(a) and 1(b), it was observed that spontaneous touching occurred less in interactions with two participants than those with three. Subsequent analysis showed that other two-participant interaction sessions not presented here also had lower rates of spontaneous touching. This contrast may be due to individual differences between participants in the study, or could be more generally linked to group sizes. To explore this question, future research could examine whether there are significant differences in social dynamics between two and three participant human-robot interactions.

III. CONCLUSIONS AND FUTURE DIRECTIONS

We have presented a case for the role of visualization in a HRI context and demonstrated through analysis of touch interactions how it can facilitate qualitative exploration of data. Resultant insights can then be verified by focused, quantitative analysis. Further data for our case study will enable statistical testing of the interaction patterns identified, informing the evolving design of our robot, particularly in terms of sensor choice and placement.

With this investigation we have taken initial steps in developing a complementary, inter-operating relationship between qualitative and quantitative research methods. Further work towards an integrated methodology will enhance our ability to reveal meaning from large quantities of data, providing a clearer picture of interaction factors for our robot via cross-method substantiation. More broadly, we envisage that visualization-driven mixed method analyses have the potential to play an important ongoing role in the discovery and communication of strategic knowledge within robot development teams, spurring new ideas and avenues of investigation.

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